Assignment – 12

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Using the operators IN, ANY, and ALL.

1) Write a query that selects all customers whose ratings are equal to or greater than ANY of Serres'.

2) Write a query using ANY or ALL that will find all salespeople who have no customers located in their city.

mysql> select Sname, city from Salespeople where city != all(sele

ct city from customers);

```
mysql> select Sname, city from Salespeople where city != all(sele
ct city from customers);
+---------+-----------+
| Sname   | city      |
+---------+-----------+
| Rifkin  | Barcelona |
| Axelord | New York  |
+---------+-----------+
2 rows in set (0.02 sec)
```

The query you've written is close, but it has a minor mistake. The != ALL condition is not appropriate here, as it will only compare the salespeople's city against every city in the customers' table, not handle the case of a salesperson having no customers in their city.

To find all salespeople who have **no customers** located in **their city**, you can use a subquery with NOT IN or ALL in a more appropriate way

mysql> select Sname, city from Salespeople as S where city != all(select city from customers as C where C.snum=S.snum);

```
mysql> select Sname, city from Salespeople as S where city != all
(select city from customers as C where C.snum=S.snum);
+---------+-----------+
| Sname   | city      |
+---------+-----------+
| Motika  | London    |
| Rifkin  | Barcelona |
| Axelord | New York  |
+---------+-----------+
3 rows in set (0.01 sec)
```

Bonus method(Not Exist Method):

SELECT Sname, city FROM Salespeople s WHERE NOT EXISTS (SELECT 1 FROM Customers c WHERE s.city = c.city AND s.Snum = c.Snum);

3) Write a query that selects all orders for amounts greater than any for the

customers in London.

mysql> select Onum,amt from orders where amt> any(select O.amt fr

om orders O inner join customers C on O.cnum=C.cnum where C.city='London');

```
mysql> select Onum,amt from orders where amt> any(select O.amt fr
om orders as O inner join customers as C on O.cnum=C.cnum where C
.city='London');
+------+---------+
| Onum | amt     |
+------+---------+
| 3002 | 1900.10 |
| 3005 | 5160.45 |
| 3006 | 1098.16 |
| 3009 | 1713.23 |
| 3008 | 4723.00 |
| 3010 | 1309.95 |
| 3011 | 9891.88 |
+------+---------+
7 rows in set (0.04 sec)
```

Note: In SQL, an **alias** is a temporary name given to a table or a column to make queries easier to write, read, and understand

In SQL, an **alias** is a temporary name given to a table or a column to make queries easier to write, read, and understand. Aliases are often used in complex queries involving multiple tables or to make column names shorter or more descriptive.

Types of Aliases:

1. **Table Alias**: A temporary name for a table.
2. **Column Alias:** A temporary name for a column in the output, which can help improve readability or change the display name in the result set.

4) Write the above query using MIN or MAX.

mysql> select Onum,amt from orders where amt>(select min(amt) from orders as O inner join customers as C on O.cnum=C.cnum where C.city='London');

```
mysql> select Onum,amt from orders where amt>(select min(amt) fro
m orders as O inner join customers as C on O.cnum=C.cnum where C.
city='London');
+------+---------+
| Onum | amt     |
+------+---------+
| 3002 | 1900.10 |
| 3005 | 5160.45 |
| 3006 | 1098.16 |
| 3009 | 1713.23 |
| 3008 | 4723.00 |
| 3010 | 1309.95 |
| 3011 | 9891.88 |
+------+---------+
7 rows in set (0.01 sec)
```